B4 Subatomic Physics — Model Solutions, 2018

Question 1 — Reactor antineutrinos

(a) Light vs. heavy stable nuclei; reactor $\bar{\nu}$

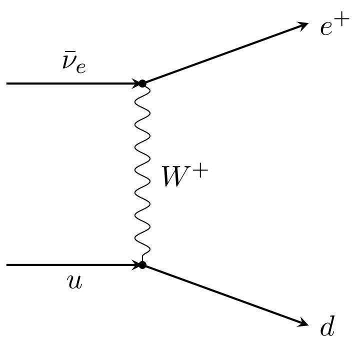
Binding from the SEMF balances surface, Coulomb, asymmetry and pairing terms. At low A, Coulomb is negligible, so the symmetric configuration $N = Z$ ($Z \approx A/2$) minimises the asymmetry term and is favoured.

For heavy nuclei the Coulomb term $\propto Z(Z-1)/A^{1/3}$ grows fast. To reduce Coulomb repulsion at fixed A, Z is shifted below $A/2$, leaving a neutron excess: stable heavy nuclei lie on the line $Z \approx A/2 - \frac{a_C}{2(a_A + a_C A^{2/3}/4)} A^{4/3}$ (qualitatively, $Z < A/2$).

Fission of a heavy nucleus produces two medium-A fragments. Inheriting the parent's neutron excess, fragments lie above the stability line for their A. They β^--decay back ($n \to p + e^- + \bar{\nu}_e$), releasing one $\bar{\nu}_e$ per β^--decay. Hence reactor flux is dominantly $\bar{\nu}_e$.

(b) Inverse beta decay diagram; energy dependence

Process: $\bar{\nu}_e + p \to e^+ + n$. At quark level, $\bar{\nu}_e + u \to e^+ + d$ via W^+ exchange (one of three u quarks in p):



Cross-section: in the reactor regime ($E_{\bar{\nu}} \sim$ few MeV $\ll M_W c^2$), the W propagator is point-like, so σ is set by phase space. With $E_{\bar{\nu}} \gg (m_n - m_p)c^2$, both outgoing particles are relativistic and

$$\sigma_{\bar{\nu}} \propto p_e E_e \approx E_{\bar{\nu}}^2.$$

Hence $\sigma_{\bar{\nu}}$ rises rapidly with $E_{\bar{\nu}}$.

(c) Why rate $\propto I$

Per target proton, the rate from $\bar{\nu}$ in $[E, E+\mathrm{d}E]$ is $\Phi(E)\sigma(E)\mathrm{d}E$, where $\Phi(E) \propto \mathrm{d}N_{\bar{\nu}}/\mathrm{d}E_{\bar{\nu}}$. Integrating over the spectrum:

$$R \propto \int \sigma_{\bar{\nu}}(E) \, \frac{\mathrm{d}N_{\bar{\nu}}}{\mathrm{d}E_{\bar{\nu}}} \, \mathrm{d}E_{\bar{\nu}} \; = \; I.$$

(d) Numerical evaluation

(i) Fission rate. Thermal power $P = 100\,\mathrm{MW} = 10^8\,\mathrm{J\,s^{-1}}$; energy per fission $E_f = 210\,\mathrm{MeV} = 210 \times 1.602 \times 10^{-13}\,\mathrm{J}$.

Compute E_f in joules:

$$E_f = 210 \times 1.602 \times 10^{-13}\,\mathrm{J} = 3.364 \times 10^{-11}\,\mathrm{J}.$$

Divide:

$$R_f = \frac{P}{E_f} = \frac{10^8}{3.364 \times 10^{-11}}\,\mathrm{s^{-1}}.$$

Evaluate:

$$\boxed{R_f \approx 2.97 \times 10^{18}\,\mathrm{fissions\,s^{-1}}.}$$

(ii) Integral I. Trapezoidal rule, $\Delta E = 1\,\mathrm{MeV}$, with $\sigma\,\mathrm{d}N/\mathrm{d}E$ from the table $\{0.04, 0.23, 0.30, 0.23, 0.04\}$ in $10^{-42}\,\mathrm{cm^2\,fission^{-1}\,MeV^{-1}}$:

$$I = \tfrac{1}{2}(0.04 + 0.04) + (0.23 + 0.30 + 0.23).$$

Sum:

$$I = 0.04 + 0.76 = 0.80\ \ [10^{-42}\,\mathrm{cm^2\,fission^{-1}}].$$

Convert to SI:

$$\boxed{I \approx 8.0 \times 10^{-47}\,\mathrm{m^2\,fission^{-1}}.}$$

(iii) Mass of scintillator. Required detection rate: $R_{\mathrm{det}} = 90/\mathrm{hr} = 90/3600 = 0.025\,\mathrm{s^{-1}}$.

Antineutrinos per unit area at distance $d = 7\,\mathrm{m}$: per fission, integrating gives $\int \mathrm{d}N/\mathrm{d}E\,\mathrm{d}E$ but the relevant flux-weighted integral is already absorbed into I. The detected rate is

$$R_{\mathrm{det}} = \frac{R_f}{4\pi d^2}\,N_p\,I,$$

where N_p is the number of free protons in the scintillator.

Solve for N_p:

$$N_p = \frac{4\pi d^2\,R_{\mathrm{det}}}{R_f\,I}.$$

Numerator:

$$4\pi d^2 R_{\mathrm{det}} = 4\pi(7)^2(0.025) = 4\pi(1.225) = 15.39\,\mathrm{m^2\,s^{-1}}.$$

Denominator:

$$R_f\,I = (2.97 \times 10^{18})(8.0 \times 10^{-47}) = 2.38 \times 10^{-28}\,\mathrm{m^2\,s^{-1}}.$$

Divide:

$$N_p = \frac{15.39}{2.38 \times 10^{-28}} = 6.47 \times 10^{28}\ \text{free protons.}$$

Free protons per C_9H_{10} monomer: 10 (the H atoms; C-bound protons are not free, but for IBD on hydrogen only H atoms count). Molar mass:

$$M = 9(12) + 10(1) = 118\,\mathrm{g\,mol^{-1}}.$$

Free protons per kg:

$$n_p = \frac{10\,N_A}{M} = \frac{10 \times 6.022 \times 10^{23}}{0.118}\,\mathrm{kg^{-1}} = 5.10 \times 10^{25}\,\mathrm{kg^{-1}}.$$

Mass needed:

$$m = \frac{N_p}{n_p} = \frac{6.47 \times 10^{28}}{5.10 \times 10^{25}}\,\mathrm{kg}.$$

Evaluate:

$$\boxed{m \approx 1.27 \times 10^3\,\mathrm{kg}\ \approx\ 1.3\,\mathrm{tonnes}.}$$

Question 2 — Decuplet baryons and the Ω^-

(a) Quark-model assignments

Decuplet members are qqq with $q \in \{u, d, s\}$ in a totally symmetric flavour–spin state.

Spin. Three spin-$\frac{1}{2}$ quarks couple to $S = \frac{1}{2}$ or $\frac{3}{2}$. The decuplet picks the $S = \frac{3}{2}$ symmetric combination, giving $J = \frac{3}{2}$.

Parity. Quarks have $P = +1$, antiquarks $P = -1$. Three quarks, $L = 0$ (ground state), so $P = (+1)^3 = +1$. Hence $J^P = \frac{3}{2}^+$.

Strangeness. $S = -N_s$, where N_s is the number of s quarks: $\Delta(0\,s) \Rightarrow S = 0$; $\Sigma^*(1\,s) \Rightarrow S = -1$; $\Xi^*(2\,s) \Rightarrow S = -2$; $\Omega^-(3\,s) \Rightarrow S = -3$.

Charges. $Q_u = +\frac{2}{3}$, $Q_d = Q_s = -\frac{1}{3}$.

State	content	Q
$\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$	uuu, uud, udd, ddd	$+2, +1, 0, -1$
$\Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}$	uus, uds, dds	$+1, 0, -1$
Ξ^{*0}, Ξ^{*-}	uss, dss	$0, -1$
Ω^-	sss	-1

Symmetry concern: sss in $S = \frac{3}{2}$ $(L = 0)$ is symmetric in space, spin, flavour. Pauli is rescued by the antisymmetric *colour* singlet — evidence for colour as a quantum number.

(b) Discovery reaction

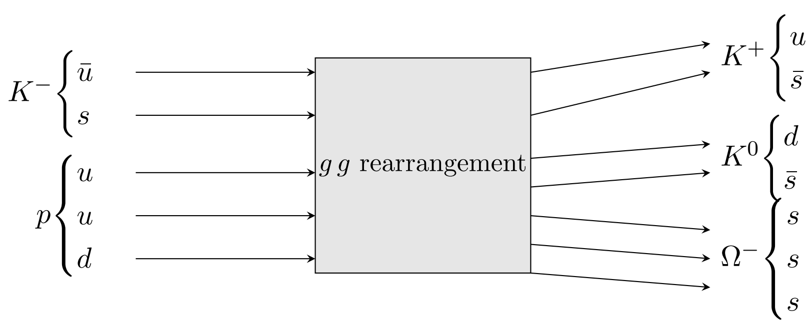
$K^-(\bar{u}s) + p(uud) \to \Omega^-(sss) + K^0(d\bar{s}) + X$.

Strangeness on LHS: $S(K^-) + S(p) = -1 + 0 = -1$. RHS: $S(\Omega^-) + S(K^0) + S(X) = -3 + 1 + S(X)$. Conservation ($\Delta S = 0$ in strong production) gives $S(X) = +1$.

Baryon number: LHS $= 1$, RHS without X has Ω^- $(B = 1)$, so $B(X) = 0$.

Charge: LHS $= -1 + 1 = 0$. RHS without X: $-1 + 0 = -1$, so $Q(X) = +1$.

Hence X is a meson with $S = +1$, $Q = +1$: $X = K^+$ $(u\bar{s})$.



(Two $s\bar{s}$ pairs are created from gluons; the original s from K^- joins them to form Ω^-. The $\bar{s}$ partners pair with u and d to form K^+ and K^0.)

(c) Mass prediction; threshold momentum

Differences along the decuplet (each step adds one s):

$$M(\Sigma^*) - M(\Delta) = 1385 - 1238 = 147\,\mathrm{MeV/c^2}.$$

$$M(\Xi^*) - M(\Sigma^*) = 1532 - 1385 = 147\,\mathrm{MeV/c^2}.$$

Linear extrapolation:

$$M(\Omega^-) \approx 1532 + 147 = 1679\,\mathrm{MeV/c^2}.$$

$$\boxed{M(\Omega^-) \approx 1679\,\mathrm{MeV/c^2}.}$$

Threshold. Lab frame: K^- on stationary p. Threshold = all final-state particles at rest in CM. Total final mass:

$$M_f = M(\Omega^-) + M(K^0) + M(K^+).$$

Use $M(K^0) \approx M(K^+) \approx 494\,\mathrm{MeV/c^2}$:

$$M_f \approx 1679 + 494 + 494 = 2667\,\mathrm{MeV/c^2}.$$

Mandelstam s at threshold:

$$s = M_f^2 c^4.$$

Lab expression:

$$s = (E_K + m_p c^2)^2 - (p_K c)^2.$$

Use $E_K^2 = (p_K c)^2 + m_K^2 c^4$ and expand:

$$s = m_K^2 c^4 + m_p^2 c^4 + 2 m_p c^2 E_K.$$

Solve for E_K:

$$E_K = \frac{M_f^2 c^4 - m_K^2 c^4 - m_p^2 c^4}{2 m_p c^2}.$$

Insert $m_K = 494$, $m_p = 938$, $M_f = 2667$ (all in MeV/c^2):

$$M_f^2 - m_K^2 - m_p^2 = 2667^2 - 494^2 - 938^2\ [\mathrm{MeV^2/c^4}].$$

Compute:

$$2667^2 = 7.113 \times 10^6; \quad 494^2 = 2.440 \times 10^5; \quad 938^2 = 8.798 \times 10^5.$$

Sum:

$$7.113 \times 10^6 - 2.440 \times 10^5 - 8.798 \times 10^5 = 5.989 \times 10^6\,\mathrm{MeV^2/c^4}.$$

Divide by $2m_p c^2 = 1876\,\mathrm{MeV}$:

$$E_K = \frac{5.989 \times 10^6}{1876}\,\mathrm{MeV} = 3192\,\mathrm{MeV}.$$

Threshold momentum:

$$p_K c = \sqrt{E_K^2 - m_K^2 c^4} = \sqrt{3192^2 - 494^2}\,\mathrm{MeV}.$$

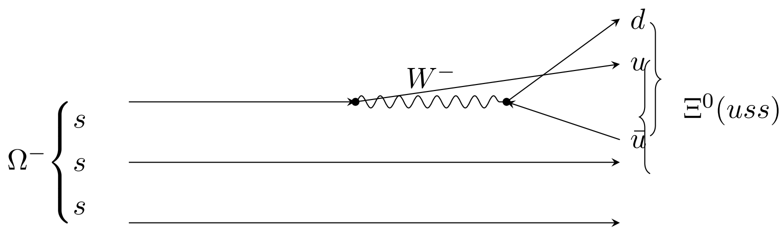
Evaluate:

$$p_K c = \sqrt{1.019 \times 10^7 - 2.44 \times 10^5} = \sqrt{9.945 \times 10^6} = 3154\,\mathrm{MeV}.$$

$$\boxed{p_{K^-}^{\min} \approx 3.15\,\mathrm{GeV/c}.}$$

(d) Decay $\Omega^- \to \Xi^0 + \pi^-$

Strangeness changes by $|\Delta S| = 1$ ($\Omega^- : S = -3 \to \Xi^0 : S = -2$): weak decay, $s \to u + W^-$, with $W^- \to d\bar{u}$.



(Spectator ss + final-state u form $\Xi^0 = uss$; $d\bar{u}$ forms π^-.)

(e) Lifetime estimate

Boost factor:

$$E_\Omega = \sqrt{(p_\Omega c)^2 + (M_\Omega c^2)^2} = \sqrt{2015^2 + 1672^2}\,\text{MeV}.$$

Compute:

$$E_\Omega = \sqrt{4.060 \times 10^6 + 2.795 \times 10^6} = \sqrt{6.856 \times 10^6} = 2618\,\text{MeV}.$$

Lorentz factors:

$$\gamma = E_\Omega/(M_\Omega c^2) = 2618/1672 = 1.566.$$

$$\beta\gamma = p_\Omega c/(M_\Omega c^2) = 2015/1672 = 1.205.$$

Lab-frame velocity:

$$\beta = (\beta\gamma)/\gamma = 1.205/1.566 = 0.770.$$

Lab speed:

$$v = \beta c = 0.770 \times 3.00 \times 10^8 = 2.31 \times 10^8\,\text{m}\,\text{s}^{-1}.$$

Lab time of flight:

$$t_{\text{lab}} = L/v = 0.025/2.31 \times 10^8 = 1.08 \times 10^{-10}\,\text{s}.$$

Proper lifetime:

$$\tau = t_{\text{lab}}/\gamma = 1.08 \times 10^{-10}/1.566.$$

$$\boxed{\tau(\Omega^-) \approx 6.9 \times 10^{-11}\,\text{s}.}$$

(Cf. PDG $\tau \approx 8.2 \times 10^{-11}\,\text{s}$ — consistent with weak decay.)

Question 3 — Baryon number, allowed reactions, GUT proton decay

(a) Baryon number assignments

- Quarks: $\mathbb{B} = +\frac{1}{3}$. Antiquarks: $\mathbb{B} = -\frac{1}{3}$.
- Leptons (and antileptons): $\mathbb{B} = 0$.
- Gauge / Higgs bosons: $\mathbb{B} = 0$.
- Baryons (qqq): $\mathbb{B} = +1$. Antibaryons ($\bar{q}\bar{q}\bar{q}$): $\mathbb{B} = -1$.
- Mesons ($q\bar{q}$): $\mathbb{B} = 0$.

(b) Allowed processes

(i) $p+p \to n+p+\pi^+$**.** $\mathbb{Q}$: $1+1 \to 0+1+1=2$. ✓ $\mathbb{B}$: $1+1 \to 1+1+0=2$. ✓ $\mathbb{L}$: $0 \to 0$. ✓ Strong/EM allowed (energy permitting). **Allowed.**

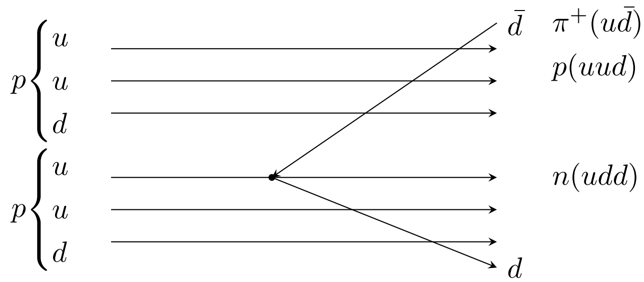
(ii) $p+e^- \to \rho^0$**.** $\mathbb{Q}$: $1-1=0 \to 0$. ✓ $\mathbb{B}$: $1+0=1 \to 0$. × $\mathbb{L}$: $0+1=1 \to 0$. × **Forbidden.**

(iii) $\mu^- \to e^- + \nu_\mu + \bar{\nu}_e$**.** $\mathbb{Q}$: $-1 \to -1$. ✓ $\mathbb{B}$: $0 \to 0$. ✓ L_μ: $1 \to 0+1+0=1$. ✓ L_e: $0 \to 1+0-1=0$. ✓ **Allowed** (standard muon decay).

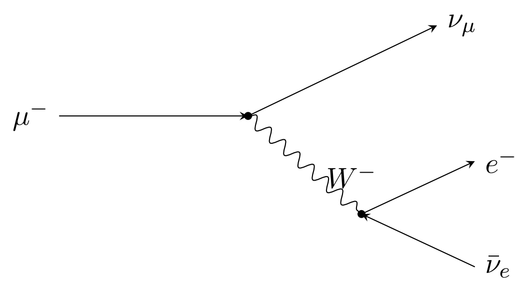
(iv) $\nu_e + e^+ \to d + \bar{u}$**.** $\mathbb{Q}$: $0+1=1 \to -\frac{1}{3}-\frac{2}{3}=-1$. × **Forbidden.**

Diagrams for allowed reactions:

(i) Quark-flow for $pp \to npp\pi^+$**:** a $u\bar{u}$ pair is created from a gluon between the protons; the new $\bar{u}$ joins one d from a proton converting that proton into $\pi^+(u\bar{d})$? — corrected: one u in proton-1 emits a gluon producing $u\bar{d}$; rearrangement gives $n(udd)+p(uud)+\pi^+(u\bar{d})$ from $p(uud)+p(uud)$.



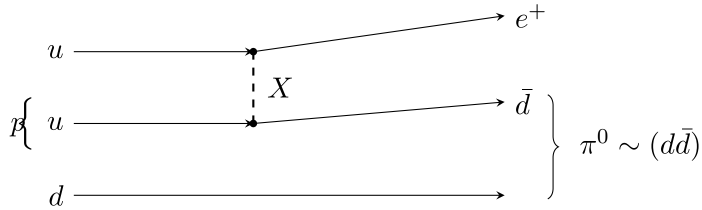
(iii) Muon decay:



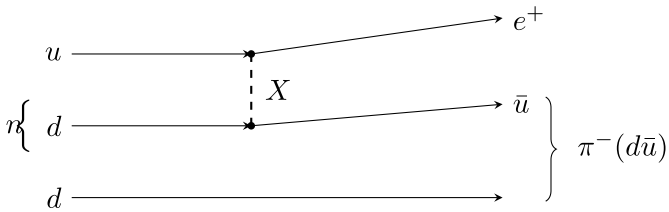
(c) GUT diagrams for $p \to e^+\pi^0$ and $n \to e^+\pi^-$

X has charge $-\frac{1}{3}$ ($e^+\bar{u}$ vertex implies $Q(X) = +1 + \frac{2}{3} = -?$; from $X \to ud$, $Q(X) = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$; problem states $|Q| = \frac{1}{3}$, so X has $Q = +\frac{1}{3}$). The two vertices $X \to ud$ (λ_q) and $X \to e^+\bar{u}$ (λ_ℓ) (and conjugates) allow $uu \to X^* \to \bar{d}\,e^+$ — equivalently $u+u \to e^+ + \bar{d}$ inside the proton.

$p(uud) \to e^+\pi^0$**:** two of the u quarks in p exchange an X, converting to $e^+ + \bar{d}$. The remaining u pairs with $\bar{d}$? — corrected vertex reading: $X \to u+d$ and $X \to e^+ + \bar{u}$. Combining: $u+\bar{d} \to X \to e^+ + \bar{u}$? rearrange: $uu \to \bar{X} \to \bar{d}\,e^+$ via $\bar{X} \to \bar{u}\bar{d}$ and $\bar{X} \to e^-u$ — equivalently $u+u \to e^+ + \bar{d}$ (crossing).



$n(udd) \to e^+\pi^-$**:** the u and one d exchange X, becoming e^+ and $\bar{d}$; remaining d pairs with $\bar{d}$ for π^-? — note $\pi^- = d\bar{u}$. Re-route: $u + d \to X^* \to e^+\bar{u}$ leaves spectator d; $\bar{u} + d \to \pi^-(d\bar{u})$.



(d) Justification of $\tau_p \propto m_X^4\hbar/(\lambda_q^2\lambda_\ell^2 m_p^5 c^2)$

The decay amplitude has two vertices, one λ_q and one λ_ℓ, joined by an X propagator at momentum transfer q^2 small compared with $m_X^2 c^2$:

$$\mathcal{M} \sim \frac{\lambda_q\lambda_\ell}{q^2 - m_X^2c^2} \approx -\frac{\lambda_q\lambda_\ell}{m_X^2c^2}.$$

Squared matrix element:

$$|\mathcal{M}|^2 \propto \frac{\lambda_q^2\lambda_\ell^2}{m_X^4c^4}.$$

The decay rate must be a rate (dimensions T^{-1}). Only mass scale available below the GUT scale is m_p. Phase space and energy release saturate at m_p, so by dimensional counting (Fermi-like effective theory):

$$\Gamma_p \sim \frac{1}{\hbar}|\mathcal{M}|^2\,(m_pc^2)^5/(\hbar c)^?.$$

Insert powers of $\hbar, c$ to make $[\Gamma] = T^{-1}$:

$$\Gamma_p = \frac{1}{A}\,\frac{\lambda_q^2\lambda_\ell^2\,m_p^5c^2}{m_X^4\,\hbar}.$$

Invert:

$$\boxed{\tau_p = \Gamma_p^{-1} = A\,\frac{m_X^4\,\hbar}{\lambda_q^2\lambda_\ell^2\,m_p^5\,c^2}.}$$

Reading the powers. The m_X^4 comes from the squared heavy-boson propagator ($\sim 1/m_X^2$ in amplitude). $\lambda_q^2\lambda_\ell^2$ from amplitude squared, two vertex factors per $\mathcal{M}$. m_p^5 comes from the three-body phase space ($\propto E^2$) times $|\mathcal{M}|^2$'s remaining mass dimension at energies $\sim m_p$ — Fermi-like rate $\propto E^5$. $\hbar/c^2$ provides the correct units.

Question 4 — Born scattering and Higgs exchange

(a) Differential cross-section, definitions of $\boldsymbol{q}$, m

$\mathrm{d}\sigma/\mathrm{d}\Omega$ = (number of particles scattered into solid angle $\mathrm{d}\Omega$ per unit time) divided by (incident flux $\times \mathrm{d}\Omega$).

In the given Born formula:

- $\boldsymbol{q} = \boldsymbol{k}' - \boldsymbol{k}$ is the momentum transfer (initial wavevector $\boldsymbol{k}$, final $\boldsymbol{k}'$). For elastic scattering, $|\boldsymbol{q}| = 2k\sin(\Theta/2)$.
- m is the (reduced) mass of the scattered particle.

(b) Yukawa cross-section; Coulomb limit

Compute the Fourier transform:

$$\tilde{V}(\boldsymbol{q}) = \int e^{i\boldsymbol{q}\cdot\boldsymbol{x}}\, V_0 \frac{e^{-\mu r}}{r}\, \mathrm{d}^3x.$$

Use spherical coordinates with $\hat{z} \parallel \boldsymbol{q}$:

$$\tilde{V}(\boldsymbol{q}) = V_0 \int_0^\infty \mathrm{d}r\, r^2 \frac{e^{-\mu r}}{r} \int_0^\pi \mathrm{d}\theta\, \sin\theta\, e^{iqr\cos\theta} \int_0^{2\pi} \mathrm{d}\phi.$$

ϕ-integral gives 2π:

$$\tilde{V}(\boldsymbol{q}) = 2\pi V_0 \int_0^\infty \mathrm{d}r\, r\, e^{-\mu r} \int_{-1}^1 \mathrm{d}u\, e^{iqru}.$$

Do the u-integral:

$$\int_{-1}^1 \mathrm{d}u\, e^{iqru} = \frac{2\sin(qr)}{qr}.$$

Insert:

$$\tilde{V}(\boldsymbol{q}) = \frac{4\pi V_0}{q} \int_0^\infty \mathrm{d}r\, e^{-\mu r} \sin(qr).$$

Use $\int_0^\infty e^{-\mu r} \sin(qr) \mathrm{d}r = q/(\mu^2 + q^2)$:

$$\tilde{V}(\boldsymbol{q}) = \frac{4\pi V_0}{q^2 + \mu^2}.$$

Substitute into Born formula:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{m}{2\pi\hbar^2}\right)^2 \frac{(4\pi V_0)^2}{(q^2+\mu^2)^2}.$$

Simplify constants:

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{A}{(q^2+\mu^2)^2}, \qquad A = \frac{4m^2V_0^2}{\hbar^4}.}$$

Coulomb limit. Coulomb potential $V(r) = Z_1Z_2e^2/(4\pi\epsilon_0 r)$ is the $\mu \to 0$ limit with $V_0 = Z_1Z_2e^2/(4\pi\epsilon_0)$:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{A}{q^4}.$$

Elastic kinematics give $|\boldsymbol{q}| = 2k\sin(\Theta/2)$:

$$q^2 = 4k^2\sin^2(\Theta/2).$$

Square:

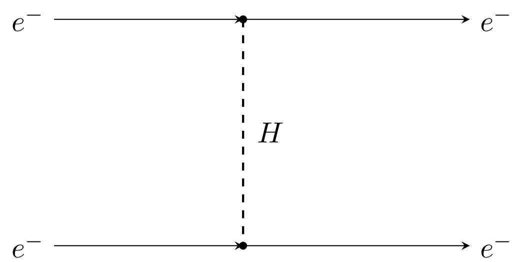
$$q^4 = 16k^4\sin^4(\Theta/2).$$

Insert:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{A}{16k^4}\frac{1}{\sin^4(\Theta/2)} = \frac{A}{16k^4}\operatorname{cosec}^4(\Theta/2).$$

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \propto \operatorname{cosec}^4(\Theta/2) \ \ \text{(Rutherford).}}$$

(c) H vs. γ exchange



In Born/perturbation theory, a single-boson exchange amplitude has matrix element of the Yukawa form $\propto g^2/(q^2+\mu^2)$ for a massive scalar; for a massless photon, $\propto g_\gamma^2/q^2$.

For the Higgs ($e\,e\,H$ coupling λ_e):

$$\mathcal{M}_H \propto \frac{\lambda_e^2}{q^2+\mu_H^2}.$$

For the photon (QED coupling $g_\gamma^2 = 4\pi\alpha$ in Heaviside–Lorentz natural units):

$$\mathcal{M}_\gamma \propto \frac{4\pi\alpha}{q^2}.$$

Take ratio:

$$R_H = \frac{\mathcal{M}_H}{\mathcal{M}_\gamma} = \frac{\lambda_e^2}{4\pi\alpha}\,\frac{q^2}{q^2+\mu_H^2}.$$

$$\boxed{R_H = \frac{\lambda_e^2}{4\pi\alpha}\,\frac{q^2}{q^2+\mu_H^2}.}$$

Maximum value at $\sqrt{s} = 200\,\mathrm{GeV}$. Maximum $|\boldsymbol{q}|$ at fixed $\sqrt{s}$: backscattering gives $q^2 = s$ for elastic massless beams (in fact $q^2_{\max} = s$ when $\Theta = \pi$, neglecting m_e):

$$q^2_{\max} = s = (200\,\mathrm{GeV})^2 = 4\times10^4\,\mathrm{GeV}^2.$$

μ_H^2:

$$\mu_H^2 = (125)^2 = 1.5625\times10^4\,\mathrm{GeV}^2.$$

Ratio of q-factors:

$$\frac{q^2}{q^2+\mu_H^2} = \frac{4\times10^4}{4\times10^4+1.5625\times10^4} = \frac{4}{5.5625} = 0.719.$$

Coupling factor:

$$\frac{\lambda_e^2}{4\pi\alpha} = \frac{(2.9\times10^{-6})^2}{4\pi\cdot(1/137)}.$$

Compute numerator:

$$\lambda_e^2 = 8.41\times10^{-12}.$$

Denominator:

$$4\pi\alpha = 4\pi/137 = 0.0917.$$

Divide:

$$\frac{\lambda_e^2}{4\pi\alpha} = \frac{8.41\times10^{-12}}{0.0917} = 9.17\times10^{-11}.$$

Multiply:

$$R_H^{\max} = 9.17 \times 10^{-11} \times 0.719.$$

$$\boxed{R_H^{\max} \approx 6.6 \times 10^{-11}.}$$

Comment. $|R_H|^2 \sim 4 \times 10^{-21}$: the Higgs exchange amplitude is suppressed by $\sim 10^{-10}$ relative to QED, and its contribution to the rate by $\sim 10^{-20}$. Even with QED interference ($\sim 2R_H$) the relative shift is $\sim 10^{-10}$ — far below experimental precision in $ee \to ee$. Direct Higgs measurement via t-channel exchange is impractical; one instead uses on-shell production (e.g. $H \to ZZ^*$, $H \to \gamma\gamma$) where σ peaks at the Higgs mass.